

PetroChina · KUNLUN Lubricant | TRANSFORMER OIL

KUNLUN Petro 45X

Naphthenic insulating oil — Inhibited (contains metal passivant)

PRODUCT DATA SHEET

Product description

Petro 45X Transformer Oil is obtained by adding a high-quality antioxidant additive to deeply-refined, low-viscosity, low-pour-point, low-sulphur-content naphthenic base oil. It offers rapid heat transfer, good oxidation stability and excellent electrical properties. The product is mainly exported and is produced under a strict quality control procedure.

Quality grade and standards

Quality grade: Highest grade.

Quality indexes according to IEC 60296:2012. Meets the Class I-30 °C specification (special purpose).

Application range

Suitable for distribution transformers, generation transformers, power transformers and reactors with a Lowest Cold Start Energizing Temperature (LCSET) of -30 °C, and oil-filled electrical equipment with similar requirements. The product contains metal passivants. Assessed by DOBLE, ABB, SIEMENS (US) and Power Grid Corporation of India.

Properties & characteristics

- Relatively low viscosity and viscosity index for effective cooling and heat transfer
- Excellent heat and oxidation stability to prolong equipment life and reduce maintenance cost
- Excellent electrical insulation: high breakdown voltage and low dielectric loss factor
- Good low-temperature properties; free of pour-point depressant; pour point below -40 °C
- Free of DBDS; passes harsh sulphur-corrosion tests (ASTM D1275B, IEC 62535, DIN 51353)
- Additives listed per IEC 60296:2012; free of polychlorinated biphenyls (PCB)

Typical properties

Quality indexes and typical data. Quality indexes according to IEC 60296:2012. Meets the Class I-30 °C specification (special purpose).

Property	Unit	Method	Min.	Max.	Typical data
Function characteristics					
Pour point	°C	ISO 3016		-40	-45
Kinematic viscosity (40 °C)	mm ² /s	ISO 3104		12	9.6
Kinematic viscosity (-30 °C)	mm ² /s	ISO 3104		1800	1650
Water content	mg/kg	IEC 60814		30	20
Breakdown voltage – untreated	kV	IEC 60156	30		40–60
Breakdown voltage – treated	kV	IEC 60156	70		>70
Density (20 °C)	kg/m ³	ISO 12185		895	886
Dielectric loss factor (90 °C)		IEC 60247		0.005	<0.001
Refining / stability characteristics					
Appearance		Visual		Clear & transparent	Clear & transparent
Acid number	mgKO	IEC 62021-1		0.01	<0.01

Property	Unit	Method	Min.	Max.	Typical data
	H/g				
Interfacial tension	mN/m	ASTM D971	40		45
Total sulphur content	%	ASTM D5453		0.15	<0.01
Corrosive sulphur		ASTM D1275B		Not corrosive	Not corrosive
Latent corrosive sulphur		IEC 62535		Not corrosive	Not corrosive
Corrosive sulphur		DIN 51353		Not corrosive	Not corrosive
DBDS content	mg/kg	IEC 62697-1		Not detectable	Not detectable
Antioxidant content	%	IEC 60666	0.08	0.40	0.38
2-Furfural & related compounds	mg/kg	IEC 61198		0.05	<0.05
Operation characteristics – Oxidation stability (120 °C, 500 h)					
Total acid number	mgKO H/g	IEC 61125C		0.3	0.08
Oil sludge	%	IEC 61125C		0.05	<0.01
Dielectric loss factor (90 °C)		IEC 60247		0.050	0.034
Health, safety and environment					
Flash point (closed)	°C	ISO 2719	135		142
PCA content	%	IP 346		3	<3
PCB content	mg/kg	IEC 61619		Not detectable	Not detectable

Note: Typical data correspond to current production and are provided for reference; they are not a specification limit. The product is produced under a strict quality control procedure with constant product resources. Transformer oil is vulnerable to moisture and contamination – keep dry and clean, store in dedicated containers indoors or in a controlled climate, and keep free of acetylene gas.